

Theory of Automata

Assignment # 1

Deadline: 12th September 2025

Instructions:

1. Attempt all questions.
2. Questions must be attempted on paper (A4 sheets or loose sheets)
3. Submit the assignment on time. Only physical submissions are accepted.

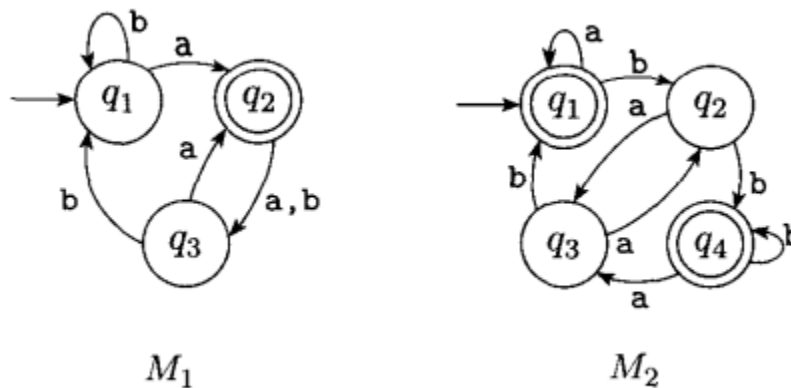
Section – 1: Regular Expressions and RegEx (1x10 + 1x3 = 13 marks)

1. Find/Create the regular expressions for the following. Alphabet set for this part is {a,b}
 - a. All strings of length greater than or equal to 3.
 - b. All strings with at least 2 a's in them.
 - c. All strings with at most 2 a's in them.
 - d. All strings with at least two occurrences of bb in them.
 - e. All strings containing even number of b's.
 - f. All string not containing bab.
 - g. All strings that start and end with different letters. (e.g. ab, ba, baa, aab)
 - h. All strings where each instance of aa must be followed by b (e.g. aab, aababaab)
 - i. All strings which contain aa or bb or both.
 - j. All strings with an even length but an odd number of a's.
2. Write RegEx for the following scenarios:
 - a. You want to define your own criteria for email IDs for your email service. The criteria are as follows:
 - i. Email ID must start with 2 numbers, followed by any word, then a + symbol, followed by another word.
 - ii. The above should always be followed by @MyEmailService.com
 - iii. It must not contain any spaces, periods or underscores or other special characters.
 - b. For your email service, in order to ensure security, you want users to create their passwords in a specific format. The format has the following rules:
 - i. Password should start with a capital English alphabet.
 - ii. Password length should be at least 14 characters.
 - iii. Password should contain a minimum of one number and one special character.

- c. Imagine there is a scrip that runs through the emails and scans them for security reasons. Write the RegEx that can be used to match the following:
- A string that has either “comp” or “imp”
 - A string that ends with “virus”
 - A string that has a “z” followed by at least one “o”
 - Examples: computer zombie virus, important zone virus, compliment zone virus, etc.

Section – 2: DFAs (5 + 1x15 = 20)

1. For the following DFAs M_1 and M_2 , answer the following questions:



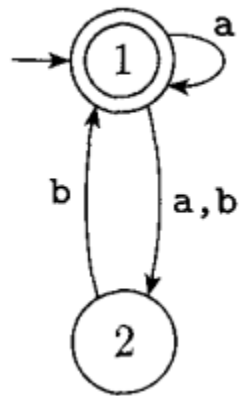
- What is the initial state?
 - What is the set of acceptance states?
 - What sequence of states does the machine go through for the following inputs:
 - $aabb$
 - $bbaa$
 - Does the machine accept the input $aabb$?
 - Does the machine accept the input ϵ ?
2. Give state diagrams of DFAs recognizing the following languages. In all parts the alphabet is $\{0,1\}$
- $\{w \mid w \text{ begins with a 1 and ends with a 0}\}$
 - $\{w \mid w \text{ contains at least three 1s}\}$
 - $\{w \mid w \text{ contains the substring 0101, i.e., } w = x0101y \text{ for some } x \text{ and } y\}$
 - $\{w \mid w \text{ has length of at least 3 and its third symbol is a 0}\}$
 - $\{w \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$
 - $\{w \mid w \text{ doesn't contain the substring 1101}\}$
 - $\{w \mid \text{the length of } w \text{ is at most 5}\}$
 - $\{w \mid \text{every odd position of } w \text{ is a 1}\}$

- i. $\{w \mid w \text{ contains at least two 0s and at most one 1}\}$
- j. $\{w \mid w \text{ contains an even number of 1s, or contains exactly two 1s}\}$
- k. The empty set
- l. All strings except the empty string
- m. $\{w \mid \text{length of } w \text{ is odd}\}$
- n. $\{w \mid \text{length of } w \text{ is even and it must not contain } 00\}$
- o. $\{w \mid \text{every third character in } w \text{ is a 1}\}$

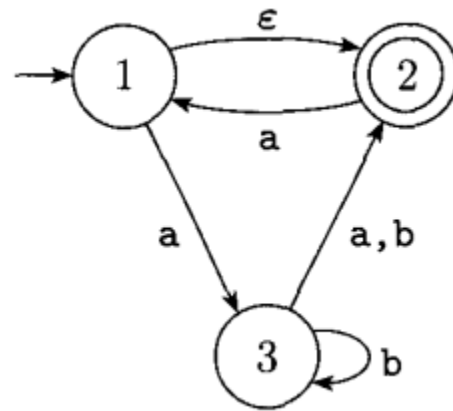
Section – 3: NFA/ ϵ -NFA to DFA Conversions (2.5x4 = 10 marks)

1. Convert the following NFAs or ϵ -NFAs to DFAs.

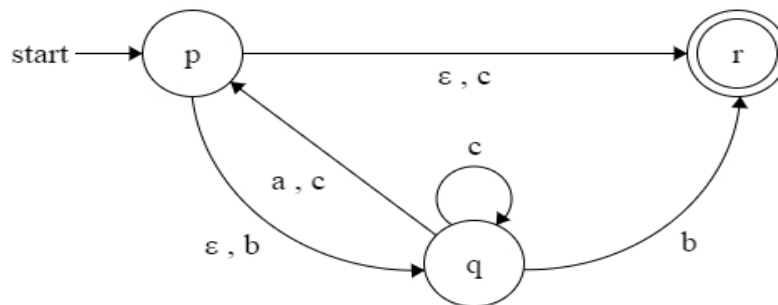
a.



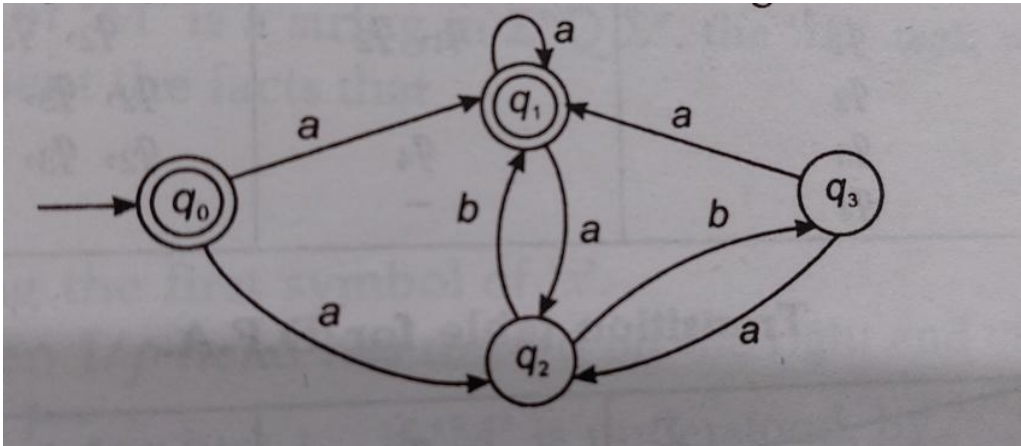
b.



c.



d.



Section – 4: DFA construction based on a Scenario (7 marks)

The ATM is a computerized machine that provides bank customers to gain access to their accounts using magnetic encoded plastic card and code number. It enables the customer to perform online transactions without involving cashier, clerk, and bank teller. The customer makes cash withdrawal, check account balances, transfer money as well as purchase prepaid mobile phone credit by using ATM card.

A typical PIN based ATM has following processes:

- Insert ATM card to establish interface. The card will be validated to ensure the correct ATM card is inserted. If the card is unsuccessfully validated, the card is ejected, and ATM session is terminated.
- For valid ATM card, user is asked to enter the PIN and press the execution key for the system to match the PIN, if PIN does not match, then user access is denied to the next stage and he or she is requested to repeat the operation with the correct PIN for a fixed two retries.
- If after the third time the PIN is unmatched, the card will be locked, and ATM session is terminated.
- If the PIN matches, then the transaction interface is displayed, and user has to select the transaction that they wish to perform.
- After the machine performs the chosen transaction, it will prompt user if a new transaction is to be performed. If the response is 'Yes', the transaction interface is again displayed and if 'No' the transaction is terminated, and card ejected, and ATM session is terminated.

Based on the above description, the states, inputs and outputs are as follows:

States:

- S1: Welcome screen
- S2: Validating card
- S3: 1st PIN entering screen
- S4: 2nd PIN entering screen
- S5: 3rd PIN entering screen.
- S6: Choosing transaction screen.
- S7: Performing transaction
- S8: Asking for other transaction screen.
- S9: Terminate ATM session

Inputs:

- A: Insert ATM card
- F: Transaction is selected
- B: Valid ATM card
- G: Complete transaction
- C: Invalid ATM card
- H: Yes, for other transaction 3 OF 4
- D: Correct PIN
- I: No for other transaction screen
- E: Incorrect PIN

Outputs:

- 0: Nothing happened
- 1: Eject ATM card
- 2: Lock ATM card

Draw the transition diagram for the system described above.